

The Sun

The Sun is a G-type main-sequence star, an ordinary yellow dwarf, and holds about 99.8 percent of the mass of the Solar System. It is about 4.6 billion years old and roughly halfway through its life on the main sequence. Its diameter is about 1.39 million kilometres, 109 times that of Earth, and about 1.3 million Earths would fit inside it.

Energy comes from nuclear fusion in the core, where hydrogen nuclei combine to form helium at a temperature of about 15 million degrees Celsius. The energy takes tens of thousands of years to work its way out to the visible surface, the photosphere, which is much cooler at about 5,500 degrees Celsius.

Light from the Sun takes about 8 minutes and 20 seconds to reach Earth. The Sun rotates faster at its equator, about 25 days per rotation, than near its poles, about 35 days, because it is not a solid body.

Sunspots are darker, cooler regions of the photosphere where strong magnetic fields suppress convection. Their number rises and falls in a cycle of about 11 years. Solar flares and coronal mass ejections are often associated with sunspot groups and can disrupt satellites and power grids on Earth.

The corona is the Sun's outer atmosphere, visible during a total solar eclipse as a pale halo. It is surprisingly hot, over a million degrees, far hotter than the surface beneath it, and why remains an active research question. The corona continuously streams charged particles outward as the solar wind, which fills the whole Solar System.

In about five billion years the Sun will exhaust the hydrogen in its core, expand into a red giant large enough to engulf Mercury and Venus, and finally shed its outer layers, leaving a dense white dwarf behind.